UC SAN DIEGO / ECE 251C

11

Wavelets from filters

Executive summary

We construct scaling functions and wavelets from their filters. The cascade algorithm repeatedly applies the refinement equation; a three-tap example shows how a box converges to a tent. Fourier analysis turns refinement into an infinite product. We then state and prove sufficient conditions under which this product defines a compactly supported scaling function with orthonormal translates and the full MRA properties. Applying the construction to Haar recovers the unit box and Euler's cosine-product identity. Finally, the conjugate-mirror filter produces the wavelet, and the discrete orthonormal basis transfers to a complete basis of $L^2(\mathbb{R})$.

11.1 Reversing the construction

So far, we have started with a scaling function and a wavelet and derived the filters that implement their analysis and synthesis operations. Lecture 10 also showed how to design an orthogonal bank directly in the sequence domain. We now consider the converse problem for functions: given a pair of filters, construct the corresponding continuous-domain scaling function and wavelet. A finite number of filter-bank stages can always be followed by another stage. Passing to infinitely many stages requires convergence, together with the orthogonality and approximation properties of an MRA.

Recall the two-scale equations:

$$\begin{aligned} \phi(t) &= \sqrt{2} \sum_{n \in \mathbb{Z}} g_0[n] \phi(2t - n), \\ \psi(t) &= \sqrt{2} \sum_{n \in \mathbb{Z}} g_1[n] \phi(2t - n). \end{aligned} \tag{11.1}$$

The first equation is a refinement equation for ϕ: the unknown function appears on both sides, at two different resolutions. Once ϕ is known, the second equation defines ψ as a finite linear combination of its dilated translates. Thus the construction begins with the low-pass filter g_0.

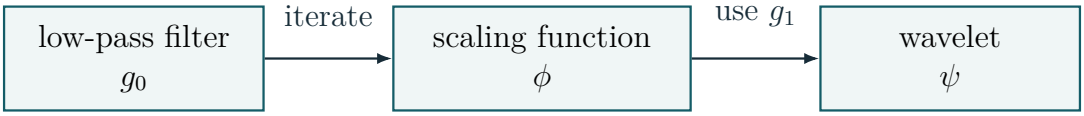


Figure 11.1: Filters determine functions through refinement. A valid finite-stage filter bank is not, by itself, a convergence theorem for infinitely many stages.

Choose an integrable, finite-energy initial function $\phi^{(0)}$ and repeatedly apply the same refinement rule:

$$\boxed{\phi^{(k+1)}(t) = \sqrt{2} \sum_{n \in \mathbb{Z}} g_0[n] \phi^{(k)}(2t - n).} \tag{11.2}$$

This iteration is the *cascade algorithm*. The superscript (k) counts refinement steps, not derivatives. Throughout this lecture the filters are FIR, so each refinement sum is finite. We use g_0 for the synthesis or refinement filter and $h_0[n] = g_0^*[-n]$ for its associated analysis filter. The two filters describe different operations and should not be interchanged.

11.2 Fixed points and the normalization of the iterates

A *fixed point* of a map F is an object x satisfying $F(x) = x$. If $x_{k+1} = F(x_k)$ converges to x and F is continuous there, then

$$F(x) = F\Big(\lim_{k \to \infty} x_k\Big) = \lim_{k \to \infty} F(x_k) = \lim_{k \to \infty} x_{k+1} = x.$$

The continuity assumption is needed to pass the limit through F. Thus convergence of an iteration identifies a fixed point only when this passage is justified. In the refinement problem, we will first verify continuity of the refinement operator and then study convergence of the iterates.

A scalar example. If $x_{k+1} = x_k/2$, then

$$x_1 = \frac{x_0}{2}, \quad x_2 = \frac{x_0}{4}, \quad x_k = \frac{x_0}{2^k} \longrightarrow 0.$$

The limit is the fixed point $0 = 0/2$. For a scaling function, we seek a nonzero fixed point of the operator

$$(\mathrm{C}_{g_0} u)(t) \coloneqq \sqrt{2} \sum_n g_0[n] u(2t - n).$$

This operator is continuous in L^2. Indeed, changing variables gives

$$\|\sqrt{2}\, u(2 \cdot - n)\|_{L^2}^2 = 2 \int_{\mathbb{R}} |u(2t - n)|^2 \, \mathrm{d}t = \int_{\mathbb{R}} |u(s)|^2 \, \mathrm{d}s.$$

The triangle inequality therefore implies

$$\|\mathrm{C}_{g_0}u - \mathrm{C}_{g_0}v\|_{L^2} \le \left(\sum_n |g_0[n]|\right) \|u - v\|_{L^2}.$$

Thus, if $\|\phi^{(k)} - \phi\|_{L^2} \to 0$, continuity gives $\mathrm{C}_{g_0}\phi = \phi$. This does not yet prove that the iterates converge.

Preserving the integral. For an integrable u, the substitution $s = 2t - n$ gives

$$\int_{\mathbb{R}} \mathrm{C}_{g_0}u(t)\,\mathrm{d}t = \frac{1}{\sqrt{2}} \left(\sum_n g_0[n]\right) \int_{\mathbb{R}} u(t)\,\mathrm{d}t.$$

Consequently, the low-pass normalization $\sum_n g_0[n] = \sqrt{2}$ preserves the integral at every refinement step. We choose an initial function with integral one. Since multiplying a refinement solution by a constant gives another solution, this normalization is needed to fix its amplitude.

11.3 A three-tap cascade: from a box toward a tent

Consider the three-tap refinement filter

$$g_0[0] = \frac{1}{2\sqrt{2}}, \qquad g_0[1] = \frac{1}{\sqrt{2}}, \qquad g_0[2] = \frac{1}{2\sqrt{2}}, \tag{11.3}$$

with all other taps zero. Its sum is $\sqrt{2}$. The cascade becomes

$$\phi^{(k+1)}(t) = \frac{1}{2}\phi^{(k)}(2t) + \phi^{(k)}(2t-1) + \frac{1}{2}\phi^{(k)}(2t-2). \tag{11.4}$$

Start with $\phi^{(0)} = \mathbf{1}_{[0,1)}$. Each new iterate is a sum of three compressed and shifted copies of the previous one.

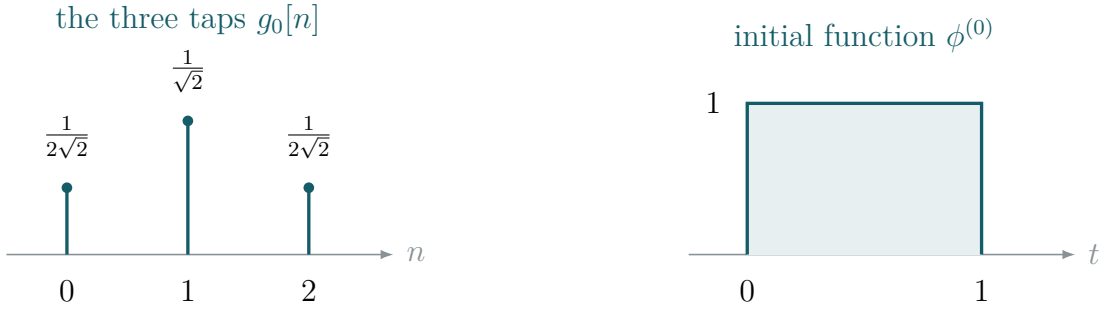


Figure 11.2: The weights in the refinement equation are $\sqrt{2}g_0 = (1/2, 1, 1/2)$. The initial box has integral one.

For the first iteration, the three supports are $[0, 1/2)$, $[1/2, 1)$, and $[1, 3/2)$. They do not overlap, so

$$\phi^{(1)}(t) = \begin{cases} \frac{1}{2}, & 0 \le t < \frac{1}{2}, \\ 1, & \frac{1}{2} \le t < 1, \\ \frac{1}{2}, & 1 \le t < \frac{3}{2}, \\ 0, & \text{otherwise.} \end{cases} \tag{11.5}$$

Its integral remains one:

$$\frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = 1.$$

Each tap specifies a weight and a shift of a half-width copy. After the first step, the function being refined is the resulting staircase rather than the original box. From the second step onward these copies overlap, so their values must be added on each interval.

The second iteration: add the overlaps. Each term in (11.4) now has cells of width $1/4$. On $[m/4, (m+1)/4)$, for $m = 0, \ldots, 6$, the resulting heights are

$$\frac{1}{4}, \quad \frac{1}{2}, \quad \frac{3}{4}, \quad 1, \quad \frac{3}{4}, \quad \frac{1}{2}, \quad \frac{1}{4}.$$

For example, on $1/2 \le t < 3/4$,

$$\phi^{(1)}(2t) = \frac{1}{2}, \qquad \phi^{(1)}(2t-1) = \frac{1}{2}, \qquad \phi^{(1)}(2t-2) = 0.$$

Thus $\phi^{(2)}(t) = \frac{1}{2}(\frac{1}{2}) + \frac{1}{2} + \frac{1}{2}(0) = \frac{3}{4}$ on that cell. The other cells follow by the same substitution.

One more iteration. The cells now have width $1/8$. From left to right, their nonzero heights are

$$\frac{1}{8}(1, 2, 3, 4, 5, 6, 7, 8, 7, 6, 5, 4, 3, 2, 1).$$

The cell heights suggest convergence to a tent on $[0, 2]$. There are two separate statements to establish: the tent satisfies the refinement equation, and the particular iterates generated from the initial box converge to it. A fixed-point identity alone does not prove the second statement.

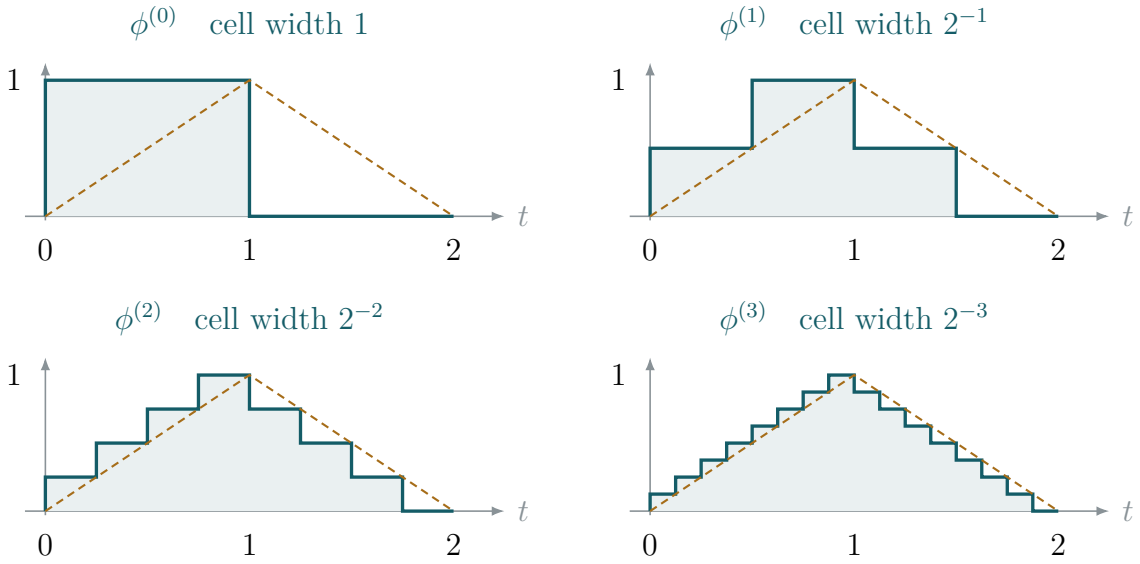


Figure 11.3: Four cascade iterates on the same axes. The dashed tent is the candidate limit. The support approaches $[0, 2]$ while the cell width halves at each step.

11.4 Checking the tent and its role in the construction

Define

$$\tau(t) = \begin{cases} t, & 0 \le t \le 1, \\ 2 - t, & 1 < t \le 2, \\ 0, & \text{otherwise.} \end{cases} \tag{11.6}$$

We verify that $\tau(t) = \frac{1}{2}\tau(2t) + \tau(2t-1) + \frac{1}{2}\tau(2t-2)$. Split its support at the breakpoints of the three terms:

t	$\frac{1}{2}\tau(2t)$	$\tau(2t-1)$	$\frac{1}{2}\tau(2t-2)$	sum
$[0, \frac{1}{2}]$	t	0	0	t
$[\frac{1}{2}, 1]$	$1-t$	$2t-1$	0	t
$[1, \frac{3}{2}]$	0	$3-2t$	$t-1$	$2-t$
$[\frac{3}{2}, 2]$	0	0	$2-t$	$2-t$

Outside $[0, 2]$, every term is zero. The formulas agree at their endpoints, so the refinement identity holds everywhere.

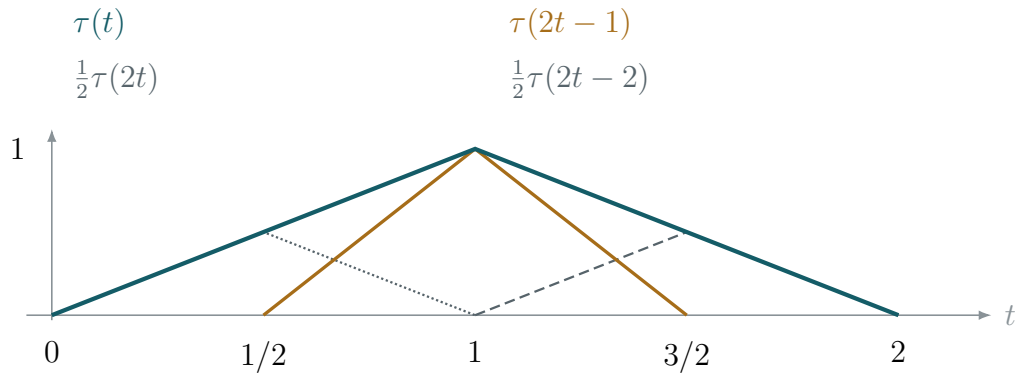


Figure 11.4: The three smaller tents add to the large tent. Where two components overlap, their slopes combine to give the slopes 1 and -1 of τ.

Refinability is not the same as orthogonality. This example illustrates the cascade, but its filter is not an orthogonal scaling filter:

$$\sum_n |g_0[n]|^2 = \frac{1}{8} + \frac{1}{2} + \frac{1}{8} = \frac{3}{4} \neq 1.$$

Adjacent translates of the tent also have a nonzero inner product:

$$\langle \tau, \tau(\cdot - 1) \rangle_{L^2} = \int_1^2 (2-t)(t-1)\,\mathrm{d}t = \int_0^1 (1-u)u\,\mathrm{d}u = \frac{1}{6}.$$

Why these particular iterates converge. Let $M = 2^k$ and write the kth iterate as

$$\phi^{(k)}(t) = \sum_m c_k[m]\mathbf{1}_{[m/M,(m+1)/M)}(t).$$

Initially $c_0[0] = 1$. Substitution in (11.4) gives

$$c_{k+1}[m] = \tfrac{1}{2}c_k[m] + c_k[m - 2^k] + \tfrac{1}{2}c_k[m - 2^{k+1}].$$

With $C_k(z) = \sum_m c_k[m]z^{-m}$, this reads

$$C_{k+1}(z) = \tfrac{1}{2}(1 + z^{-2^k})^2 C_k(z), \qquad C_0(z) = 1.$$

Iterating, and setting $q = z^{-1}$, yields

$$C_k(z) = 2^{-k}\left[\prod_{r=0}^{k-1}(1 + q^{2^r})\right]^2.$$

Repeated difference of squares gives

$$(1-q)\prod_{r=0}^{k-1}(1 + q^{2^r}) = 1 - q^{2^k}.$$

Comparing with $(1-q)\sum_{r=0}^{M-1} q^r = 1-q^M$ proves the polynomial identity

$$C_k(z) = \frac{1}{M}\left(\sum_{r=0}^{M-1} z^{-r}\right)^2.$$

The coefficient of z^{-m} counts pairs of integers in $\{0,\dots,M-1\}$ whose sum is m. There are $m+1$ such pairs for $0 \le m < M$, and $2M-1-m$ for $M \le m \le 2M-2$. Thus

$$c_k[m] = \begin{cases} (m+1)/M, & 0 \le m < M, \\ (2M-1-m)/M, & M \le m \le 2M-2, \\ 0, & \text{otherwise.} \end{cases}$$

On every cell meeting $[0,2]$, including the final zero-height cell, this height equals the tent at the right endpoint. Since the tent changes by at most the cell width,

$$|\phi^{(k)}(t) - \tau(t)| \le 2^{-k}, \qquad \|\phi^{(k)} - \tau\|_{L^2}^2 \le 2\,4^{-k} \longrightarrow 0.$$

The second bound follows by integrating over $[0,2]$, outside which both functions vanish. This proves convergence, not just the fixed-point identity.

11.5 Fourier analysis turns refinement into multiplication

The tent example allowed us to compute each iterate explicitly. For a general filter, we need a description that does not require tracking every interval separately. The cascade combines shifted, compressed functions, and Fourier analysis separates those operations into a phase factor and a change of frequency. Assume first that ϕ is integrable, and recall

$$\widehat{\phi}(\omega) = \int_{\mathbb{R}} \phi(t) \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t, \qquad G_0(z) = \sum_n g_0[n] z^{-n}.$$

The first expression transforms a function on $\mathbb{R}$; evaluating the second on $z = \mathrm{e}^{\mathrm{j}\theta}$ gives the DTFT of a sequence.

Transform the refinement equation term by term:

$$\widehat{\phi}(\omega) = \sqrt{2}\sum_n g_0[n] \int_{-\infty}^{\infty} \phi(2t-n)\mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t.$$

For each fixed n, substitute

$$u = 2t - n, \qquad t = \frac{u+n}{2}, \qquad \mathrm{d}t = \frac{1}{2}\,\mathrm{d}u.$$

The limits remain $-\infty$ and $+\infty$, because the slope is positive. Hence

$$\begin{aligned}\widehat{\phi}(\omega) &= \frac{1}{\sqrt{2}} \sum_n g_0[n] \int_{-\infty}^{\infty} \phi(u) \mathrm{e}^{-\mathrm{j}\omega(u+n)/2} \, \mathrm{d}u \\ &= \frac{1}{\sqrt{2}} \left(\sum_n g_0[n] \mathrm{e}^{-\mathrm{j}(\omega/2)n} \right) \left(\int_{\mathbb{R}} \phi(u) \mathrm{e}^{-\mathrm{j}(\omega/2)u} \, \mathrm{d}u \right) \\ &= \frac{G_0(\mathrm{e}^{\mathrm{j}\omega/2})}{\sqrt{2}} \, \widehat{\phi}(\omega/2).\end{aligned}$$

Define the normalized response

$$m_0(\theta) \coloneqq \frac{G_0(\mathrm{e}^{\mathrm{j}\theta})}{\sqrt{2}}. \tag{11.7}$$

Then

$$\boxed{\widehat{\phi}(\omega) = m_0(\omega/2)\widehat{\phi}(\omega/2).} \tag{11.8}$$

The argument of m_0 is $\omega/2$, not ω. The same compression that replaces t by $2t$ replaces the Fourier argument by $\omega/2$.

11.6 The infinite-product formula

Apply (11.8) repeatedly:

$$\begin{aligned}\widehat{\phi}(\omega) &= m_0(\omega/2)\widehat{\phi}(\omega/2) \\ &= m_0(\omega/2)m_0(\omega/4)\widehat{\phi}(\omega/4) \\ &= m_0(\omega/2)m_0(\omega/4)m_0(\omega/8)\widehat{\phi}(\omega/8).\end{aligned}$$

For a positive integer I, put

$$\mathcal{P}_I(\omega) \coloneqq \prod_{i=1}^{I} m_0(\omega/2^i).$$

The exact finite identity is

$$\widehat{\phi}(\omega) = \mathcal{P}_I(\omega)\widehat{\phi}(\omega/2^I). \tag{11.9}$$

Normalize $\int_{\mathbb{R}} \phi(t) \, \mathrm{d}t = \widehat{\phi}(0) = 1$. The Fourier transform of an integrable function is continuous, so for fixed ω,

$$\widehat{\phi}(\omega/2^I) \longrightarrow 1.$$

In particular, this factor is nonzero for all sufficiently large I. Dividing (11.9) by it and taking the limit gives

$$\boxed{\widehat{\phi}(\omega) = \prod_{i=1}^{\infty} \frac{G_0(\mathrm{e}^{\mathrm{j}\omega/2^i})}{\sqrt{2}}.} \tag{11.10}$$

Thus, whenever a normalized integrable refinement solution exists, its Fourier transform is determined by g_0. Uniqueness of the Fourier transform then determines the function itself, up to changes on sets of measure zero. The product formula therefore gives uniqueness; the existence and convergence questions still require an argument.

The actual cascade retains its initial function. Applying the same Fourier calculation to (11.2) gives

$$\widehat{\phi}^{(I)}(\omega) = \mathcal{P}_I(\omega)\widehat{\phi}^{(0)}(\omega/2^I). \tag{11.11}$$

The factor $\widehat{\phi}^{(0)}(\omega/2^I)$ retains the dependence on the initial function. For an integrable initial function of integral one, this factor tends to one at every fixed frequency. This is not yet enough to conclude that the cascade converges in L^2: the error must be controlled over the entire frequency axis, not just at each fixed frequency. By Plancherel,

$$\|\phi^{(I)} - \phi\|_{L^2}^2 = \frac{1}{2\pi}\int_{\mathbb{R}} |\widehat{\phi}^{(I)}(\omega) - \widehat{\phi}(\omega)|^2 \,\mathrm{d}\omega.$$

An infinite-scale construction must control this full energy. Convergence also depends on the initial function, not only on the filter.

11.7 The fundamental theorem of wavelet analysis

The infinite product suggests a candidate for the Fourier transform of the scaling function. We next give sufficient conditions under which this candidate defines an orthonormal MRA. It is important to distinguish the filter identities that every such construction must satisfy from the additional condition that will allow us to prove existence.

The following result is the fundamental theorem of wavelet analysis in the Mallat–Meyer framework. The convergence criterion is due to Mallat (1989). For FIR filters, the regularity of the frequency response is automatic; the substantive extra hypothesis is its nonvanishing on the middle half of a frequency period.

Theorem 11.1 (Mallat–Meyer: necessary conditions and a sufficient construction)**.** Suppose an integrable scaling function ϕ generates an orthonormal MRA, has $\int_{\mathbb{R}} \phi(t)\,\mathrm{d}t = 1$, and has an FIR refinement filter g_0. Then

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2 \quad (\omega \in \mathbb{R}), \tag{11.12}$$

$$G_0(1) = \sum_n g_0[n] = \sqrt{2}. \tag{11.13}$$

Conversely, let an FIR filter g_0 satisfy these two conditions. If, in addition,

$$\boxed{\min_{|\omega|\le\pi/2} |G_0(\mathrm{e}^{\mathrm{j}\omega})| > 0,} \tag{11.14}$$

then the infinite product in (11.10) is the Fourier transform of a compactly supported scaling function $\phi \in L^2(\mathbb{R})$ with integral one. Its integer translates are orthonormal, and the closed spaces

$$V_i = \overline{\operatorname{span}\{2^{i/2}\phi(2^i\,\cdot - n) : n \in \mathbb{Z}\}}^{L^2}, \qquad i \in \mathbb{Z},$$

form an orthonormal multiresolution analysis: they are nested, satisfy the scaling and translation rules, and obey

$$\overline{\bigcup_{i\in\mathbb{Z}} V_i}^{L^2} = L^2(\mathbb{R}), \qquad \bigcap_{i\in\mathbb{Z}} V_i = \{0\}.$$

For an FIR filter, the response is a finite sum of exponentials, so all of its derivatives exist and are continuous. The additional nonvanishing assumption will keep the infinite product away from zero on a fixed neighborhood of the origin. This is what allows the pointwise product calculation to be strengthened to an energy-convergence argument.

> **The two directions have different assumptions**
> Power complementarity and the gain $\sqrt{2}$ are necessary. The nonvanishing condition on $[-\pi/2, \pi/2]$ is an *additional sufficient condition* for the converse, not a necessary condition on every orthonormal scaling filter.

The theorem guarantees an MRA, not merely a formal solution of the refinement equation. This is precisely what the tent example did not guarantee.

Why the necessary conditions hold. In the orthonormal basis $\phi_{1,n}(t) = \sqrt{2}\phi(2t-n)$ of V_1, refinement gives

$$\phi(t) = \sum_n g_0[n]\phi_{1,n}(t), \qquad \phi(t-k) = \sum_n g_0[n-2k]\phi_{1,n}(t).$$

For the second identity, replace t by $t-k$ in the first and reindex the sum. Taking the inner product, conjugating the coefficients in its second argument, yields

$$\begin{aligned}\delta[k] &= \langle \phi, \phi(\cdot - k)\rangle_{L^2} \\ &= \sum_{n,m} g_0[n] g_0^*[m-2k]\langle \phi_{1,n}, \phi_{1,m}\rangle_{L^2} \\ &= \sum_n g_0[n] g_0^*[n-2k].\end{aligned}$$

To recover the power condition directly, expand the squared DTFT and keep even differences of indices:

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 + |G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 2\sum_k \left(\sum_n g_0[n] g_0^*[n-2k]\right) \mathrm{e}^{-2\mathrm{j}k\omega} = 2.$$

Integrating refinement also gives

$$1 = \sqrt{2}\sum_n g_0[n] \int_{\mathbb{R}} \phi(2t-n)\,\mathrm{d}t = \frac{1}{\sqrt{2}}\sum_n g_0[n],$$

which proves (11.13).

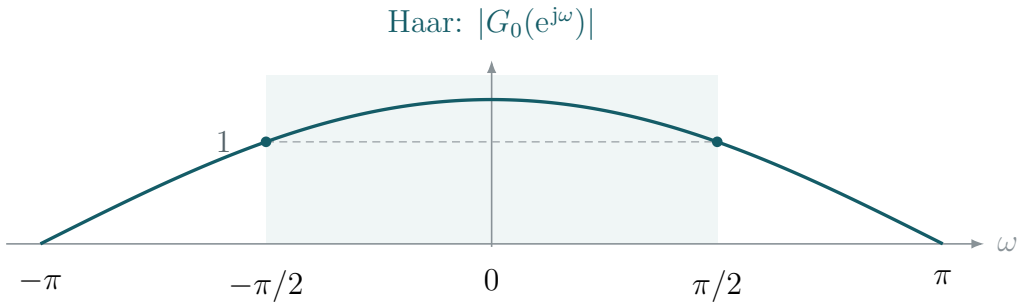


Figure 11.5: The nonvanishing interval is the shaded middle half of one frequency period. For Haar the magnitude is at least one there; zeros at $\pm\pi$ do not violate the condition.

Structure of the converse proof. We first establish convergence of the product and obtain a lower bound near the origin. We then use power complementarity to control the energy of suitable finite products and prove orthonormality of the limiting translates. Finally, we connect this limit to the box-seeded cascade and verify the two limiting properties of an MRA. These steps are kept separate because none follows from pointwise product convergence alone.

Converse, step 1: the product exists and stays nonzero near zero. Write

$$m_0(\omega) = G_0(\mathrm{e}^{\mathrm{j}\omega})/\sqrt{2}, \qquad \mathcal{P}_I(\omega) = \prod_{i=1}^{I} m_0(\omega/2^i), \qquad \mathcal{P}_0 = 1.$$

The assumptions give $m_0(0) = 1$ and

$$|m_0(\omega)|^2 + |m_0(\omega+\pi)|^2 = 1, \qquad |m_0(\omega)| \le 1.$$

Because the sum defining m_0 is finite, its derivative is bounded. In particular, with $L = \frac{1}{\sqrt{2}}\sum_n |n g_0[n]|$, integration of the derivative gives $|m_0(s) - 1| \le L|s|$.

The identity $\prod_{i=1}^{I+1} a_i - \prod_{i=1}^{I} a_i = (a_{I+1} - 1)\prod_{i=1}^{I} a_i$ telescopes. Since every factor has magnitude at most one, for $R > I$,

$$|\mathcal{P}_R(\omega) - \mathcal{P}_I(\omega)| \le \sum_{i=I+1}^{R} |m_0(\omega/2^i) - 1| \le L|\omega| \sum_{i=I+1}^{\infty} 2^{-i} = L|\omega|2^{-I}.$$

On every bounded frequency interval this bound tends to zero uniformly. Thus the products converge uniformly there to a continuous function

$$\Phi(\omega) \coloneqq \lim_{I\to\infty} \mathcal{P}_I(\omega), \qquad \Phi(0) = 1, \qquad |\Phi(\omega)| \le 1.$$

Separating the first I factors from the tail gives the exact identity

$$\Phi(\omega) = \mathcal{P}_I(\omega)\Phi(\omega/2^I).$$

We also need a positive lower bound on $[-\pi, \pi]$. Let $c = \min_{|s|\le\pi/2} |m_0(s)| > 0$ and choose R so that $L\pi 2^{-R} \le 1/2$. The same telescoping estimate, applied to the tail alone, gives

$$\left| \prod_{i=R+1}^{\infty} m_0(\omega/2^i) - 1 \right| \le L\pi 2^{-R} \le \tfrac{1}{2} \qquad (|\omega| \le \pi).$$

The first R factors have magnitude at least c, and the tail has magnitude at least $1/2$. Consequently,

$$\boxed{|\Phi(\omega)| \ge b \coloneqq \tfrac{1}{2}c^R > 0 \qquad (|\omega| \le \pi).}$$

The lower bound prevents the low-frequency part of the limiting product from degenerating. In the next steps it will provide an integrable bound for the expanding finite products, which is needed to control their frequency-domain tails.

Step 2: finite products with controlled energy. Although $\mathcal{P}_I$ is bounded, it is periodic and does not belong to $L^2(\mathbb{R})$. We therefore restrict it to an interval that grows with I. This produces finite-energy functions without changing their values on any fixed bounded interval once I is sufficiently large:

$$F_I(\omega) \coloneqq \mathcal{P}_I(\omega)\mathbf{1}_{[-2^I\pi,\, 2^I\pi)}(\omega), \qquad A_I[k] \coloneqq \int_{\mathbb{R}} |F_I(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega}\, \mathrm{d}\omega.$$

Here $k \in \mathbb{Z}$. We will show that $A_I[k] = 2\pi\delta[k]$ for every $I \ge 0$.

For $I \ge 1$, put $T = 2^I\pi$. Split the integral into $[-T, 0)$ and $[0, T)$, and set $v = \omega + T$ in the first part. If $i < I$, then $T/2^i = 2^{I-i}\pi$ is a multiple of 2π. Thus

$$\mathcal{P}_{I-1}(v - T) = \mathcal{P}_{I-1}(v), \qquad \mathrm{e}^{\mathrm{j}k(v-T)} = \mathrm{e}^{\mathrm{j}kv}.$$

The last factor has its argument shifted by π. Combining the two integrals therefore gives

$$\begin{aligned} A_I[k] &= \int_0^T |\mathcal{P}_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv} \left(|m_0(v/2^I - \pi)|^2 + |m_0(v/2^I)|^2 \right) \mathrm{d}v \\ &= \int_0^T |\mathcal{P}_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv}\, \mathrm{d}v. \end{aligned}$$

The expression in parentheses equals one by power complementarity and periodicity. The remaining integrand has period T, so its integral over $[0, T)$ equals its integral over $[-T/2, T/2)$:

$$A_I[k] = \int_{-2^{I-1}\pi}^{2^{I-1}\pi} |\mathcal{P}_{I-1}(v)|^2 \mathrm{e}^{\mathrm{j}kv}\,\mathrm{d}v = A_{I-1}[k].$$

At $I = 0$, the product is one. Therefore

$$A_0[k] = \int_{-\pi}^{\pi} \mathrm{e}^{\mathrm{j}k\omega}\,\mathrm{d}\omega = \begin{cases} 2\pi, & k = 0, \\ 0, & k \neq 0, \end{cases}$$

where the second case follows by integrating the exponential and using $\mathrm{e}^{\mathrm{j}k\pi} = \mathrm{e}^{-\mathrm{j}k\pi} = (-1)^k$. Induction proves

$$\boxed{A_I[k] = 2\pi\delta[k], \qquad \|F_I\|_{L^2}^2 = 2\pi.}$$

The expanding frequency window has constant total energy. We must still show that none of this energy escapes to arbitrarily high frequencies as I grows.

Step 3: convergence in energy and orthonormal translates. For any fixed $B > 0$, uniform convergence on $[-B, B]$ gives

$$\int_{-B}^{B} |\Phi(\omega)|^2\,\mathrm{d}\omega = \lim_{I\to\infty} \int_{-B}^{B} |F_I(\omega)|^2\,\mathrm{d}\omega \leq 2\pi.$$

Letting B increase shows that $\Phi \in L^2(\mathbb{R})$ and $\|\Phi\|_{L^2}^2 \leq 2\pi$. Inside the support of F_I, the identity from step 1 and its lower bound give

$$|\Phi(\omega)| = |\mathcal{P}_I(\omega)|\,|\Phi(\omega/2^I)| \geq b|F_I(\omega)|.$$

Outside that support $F_I = 0$, so the inequality $|F_I| \leq b^{-1}|\Phi|$ holds everywhere. Hence

$$|F_I - \Phi|^2 \leq 2(1 + b^{-2})|\Phi|^2.$$

This bound makes the convergence in energy explicit. On $[-B, B]$, the integral of $|F_I - \Phi|^2$ tends to zero by uniform convergence. Outside this interval it is at most

$$2(1 + b^{-2}) \int_{|\omega|>B} |\Phi(\omega)|^2\,\mathrm{d}\omega,$$

which can be made arbitrarily small independently of I. First choose B, then I. This proves $\|F_I - \Phi\|_{L^2} \to 0$, and therefore $\|\Phi\|_{L^2}^2 = 2\pi$.

Plancherel gives a function $\phi \in L^2(\mathbb{R})$ with Fourier transform Φ and $\|\phi\|_{L^2} = 1$. Explicitly, inverse Fourier transformation means

$$\frac{1}{2\pi} \int_{-B}^{B} \Phi(\omega)\mathrm{e}^{\mathrm{j}\omega t}\,\mathrm{d}\omega \xrightarrow[B\to\infty]{L^2} \phi(t).$$

Moreover, Cauchy–Schwarz gives

$$\int_{\mathbb{R}} \big||F_I|^2 - |\Phi|^2\big|\,\mathrm{d}\omega \le \|F_I - \Phi\|_{L^2}\big(\|F_I\|_{L^2} + \|\Phi\|_{L^2}\big) \longrightarrow 0.$$

We may therefore pass to the limit in $A_I[k]$. Using the inner product $\langle f, g\rangle_{L^2} = \int f(t)g(t)^*\,\mathrm{d}t$,

$$\langle \phi, \phi(\cdot - k)\rangle_{L^2} = \frac{1}{2\pi}\int_{\mathbb{R}} |\Phi(\omega)|^2 \mathrm{e}^{\mathrm{j}k\omega}\,\mathrm{d}\omega = \delta[k].$$

Thus the integer translates of ϕ are orthonormal.

Step 4: the box-seeded cascade converges to this function. Let $u_0 = \mathbf{1}_{[0,1)}$ and $u_{I+1} = \mathrm{C}_{g_0} u_I$. Power complementarity implies

$$\sum_n g_0[n] g_0^*[n - 2k] = \delta[k].$$

Indeed, expanding the power identity gives $1 = \sum_k (\sum_n g_0[n] g_0^*[n-2k])\mathrm{e}^{-2\mathrm{j}k\omega}$; multiplication by $\mathrm{e}^{2\mathrm{j}r\omega}$ and integration over $[-\pi, \pi]$ extracts the coefficient $\delta[r]$.

The integer translates of u_0 are orthonormal, since the unit cells are disjoint. If the translates of u_I are orthonormal, then so are $\{\sqrt{2}u_I(2\,\cdot - n)\}_{n\in\mathbb{Z}}$. Expanding the next refinement gives

$$\begin{aligned}\langle u_{I+1}, u_{I+1}(\cdot - k)\rangle_{L^2} &= \sum_{n,r} g_0[n] g_0^*[r - 2k] \\ &\qquad \times \langle \sqrt{2}u_I(2\,\cdot - n), \sqrt{2}u_I(2\,\cdot - r)\rangle_{L^2} \\ &= \sum_n g_0[n] g_0^*[n - 2k] = \delta[k].\end{aligned}$$

Induction shows $\|u_I\|_{L^2} = 1$ at every stage.

Write $U_I = \widehat{u}_I$. The cascade formula gives

$$U_I(\omega) = \mathcal{P}_I(\omega)\widehat{u}_0(\omega/2^I), \qquad \widehat{u}_0(s) = \int_0^1 \mathrm{e}^{-\mathrm{j}st}\,\mathrm{d}t.$$

Since $|\mathrm{e}^{-\mathrm{j}st} - 1| \le |s|t$, we have $|\widehat{u}_0(s) - 1| \le |s|/2$. Thus $U_I \to \Phi$ uniformly on bounded intervals. Both transforms have squared L^2 norm 2π.

To control their inner product, split the integral at B. On $[-B, B]$, uniform convergence gives $\int U_I \Phi^* \to \int |\Phi|^2$. The remaining part satisfies

$$\left|\int_{|\omega|>B} U_I \Phi^*\,\mathrm{d}\omega\right| \le \sqrt{2\pi}\left(\int_{|\omega|>B} |\Phi|^2\,\mathrm{d}\omega\right)^{1/2},$$

which tends to zero as $B \to \infty$, uniformly in I. Consequently $\langle u_I, \phi\rangle_{L^2} \to 1$, and

$$\boxed{\|u_I - \phi\|_{L^2}^2 = 2 - 2\,\mathrm{Re}\langle u_I, \phi\rangle_{L^2} \longrightarrow 0.}$$

This proves convergence for the specified unit-box seed, not for an arbitrary initial function.

Step 5: compact support, normalization, and nested spaces. Choose integers $a < b$ containing the support of g_0. Refinement sends a function supported in $[\alpha, \beta]$ to one supported in $[(\alpha + a)/2, (\beta + b)/2]$. Starting from $[0, 1]$, induction gives

$$\operatorname{supp} u_I \subset [a(1 - 2^{-I}),\ b + (1 - b)2^{-I}].$$

For every $\varepsilon > 0$, these supports eventually lie in $[a - \varepsilon, b + \varepsilon]$. Since $u_I \to \phi$ in L^2, the integral of $|\phi|^2$ outside that interval is zero. Letting $\varepsilon \downarrow 0$ shows $\operatorname{supp} \phi \subset [a, b]$.

All iterates and the limit lie in the fixed bounded interval $D = [\min(0, a), \max(1, b)]$. Cauchy–Schwarz gives

$$\|u_I - \phi\|_{L^1} \le |D|^{1/2} \|u_I - \phi\|_{L^2} \longrightarrow 0.$$

Their integrals are one, so $\int \phi = 1$. This also shows that the ordinary Fourier integral of ϕ equals the continuous product Φ everywhere. Continuity of C_{g_0}, proved in Section 11.2, now yields $\mathrm{C}_{g_0}\phi = \phi$.

Define $\phi_{i,n}(t) = 2^{i/2}\phi(2^i t - n)$ and $V_i = \overline{\operatorname{span}\{\phi_{i,n} : n \in \mathbb{Z}\}}^{L^2}$. Changing variables gives

$$\langle \phi_{i,n}, \phi_{i,r} \rangle_{L^2} = \langle \phi(\cdot - n), \phi(\cdot - r) \rangle_{L^2} = \delta[n - r].$$

Thus these functions are an orthonormal basis of their closed span V_i. Refinement, followed by $r = n + 2k$, gives

$$\begin{aligned} \phi_{i,k}(t) &= \sqrt{2}\, 2^{i/2} \sum_n g_0[n] \phi(2^{i+1} t - 2k - n) \\ &= \sum_r g_0[r - 2k] \phi_{i+1,r}(t). \end{aligned}$$

Every basis function of V_i is in V_{i+1}, hence $V_i \subset V_{i+1}$.

Translation by $2^{-i}k$ replaces $\phi_{i,n}$ by $\phi_{i,n+k}$, so V_i is invariant under these translations. The norm-preserving map $f(t) \mapsto \sqrt{2} f(2t)$ takes $\phi_{i,n}$ to $\phi_{i+1,n}$ and maps V_i onto V_{i+1}. These are the translation and scaling rules in the MRA definition. The two limiting properties remain to be checked.

Step 6: fine resolutions approximate every finite-energy signal. Let P_i denote the orthogonal projection onto V_i:

$$P_i f = \sum_n \langle f, \phi_{i,n} \rangle_{L^2} \phi_{i,n}, \qquad \|f - P_i f\|_{L^2}^2 = \|f\|_{L^2}^2 - \sum_n |\langle f, \phi_{i,n} \rangle_{L^2}|^2.$$

This is the basis projection identity; it also gives $\|P_i f\|_{L^2} \le \|f\|_{L^2}$. First suppose $\widehat{f}$ vanishes outside $[-B, B]$, and take i large enough that $2^i\pi > B$. The scaling and translation formula gives

$$\widehat{\phi}_{i,n}(\omega) = 2^{-i/2}\mathrm{e}^{-\mathrm{j}\omega n/2^i}\Phi(\omega/2^i).$$

Parseval followed by $\theta = \omega/2^i$ therefore yields

$$\begin{aligned}\langle f, \phi_{i,n}\rangle_{L^2} &= \frac{2^{-i/2}}{2\pi}\int_{\mathbb{R}} \widehat{f}(\omega)\Phi(\omega/2^i)^*\mathrm{e}^{\mathrm{j}\omega n/2^i}\,\mathrm{d}\omega \\ &= \frac{2^{i/2}}{2\pi}\int_{-\pi}^{\pi} \widehat{f}(2^i\theta)\Phi(\theta)^*\mathrm{e}^{\mathrm{j}n\theta}\,\mathrm{d}\theta.\end{aligned}$$

These are Fourier-series coefficients, up to the factor $2^{i/2}$ and a sign of the index. The Fourier-series Parseval identity is

$$\sum_n \left|\frac{1}{2\pi}\int_{-\pi}^{\pi} F(\theta)\mathrm{e}^{\mathrm{j}n\theta}\,\mathrm{d}\theta\right|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi}|F(\theta)|^2\,\mathrm{d}\theta.$$

Applying it and changing variables back gives

$$\|P_i f\|_{L^2}^2 = \frac{1}{2\pi}\int_{-B}^{B}|\widehat{f}(\omega)|^2|\Phi(\omega/2^i)|^2\,\mathrm{d}\omega.$$

Because Φ is continuous at zero with $\Phi(0) = 1$, the factor $|\Phi(\omega/2^i)|^2$ tends uniformly to one on $[-B, B]$. Thus $\|P_i f\|_{L^2}^2 \to \|f\|_{L^2}^2$, so $P_i f \to f$.

For general $f \in L^2$, let f_B have transform $\widehat{f}\mathbf{1}_{[-B,B]}$. Plancherel gives $\|f - f_B\|_{L^2} \to 0$. Since projection cannot increase the norm,

$$\|f - P_i f\|_{L^2} \le 2\|f - f_B\|_{L^2} + \|f_B - P_i f_B\|_{L^2}.$$

Choose B first and then i. Hence $P_i f \to f$ for every f, proving that the union of the spaces V_i is dense in L^2.

Step 7: arbitrarily coarse resolutions retain only zero. First take f supported in $[-B, B]$. Cauchy–Schwarz on that interval gives

$$|\langle f, \phi_{i,n}\rangle_{L^2}|^2 \le \|f\|_{L^2}^2\int_{-B}^{B} 2^i|\phi(2^i t - n)|^2\,\mathrm{d}t = \|f\|_{L^2}^2\int_{-2^iB-n}^{2^iB-n}|\phi(u)|^2\,\mathrm{d}u.$$

For sufficiently negative i, these intervals are disjoint. Because ϕ is supported in $[a, b]$, at most $b - a + 3$ of them meet its support. If E_i is their union intersected with $[a, b]$, then

$$\|P_i f\|_{L^2}^2 \le \|f\|_{L^2}^2\int_{E_i}|\phi(u)|^2\,\mathrm{d}u, \qquad |E_i| \le (b - a + 3)\,2^{i+1}B \longrightarrow 0.$$

To see why the integral vanishes, fix $T > 0$ and write

$$\int_{E_i}|\phi|^2 \le T|E_i| + \int_{\{|\phi|^2 > T\}}|\phi|^2.$$

The second term is arbitrarily small for large T, since $|\phi|^2$ is integrable. With T fixed, the first term tends to zero. Therefore $P_i f \to 0$ as $i \to -\infty$.

For arbitrary f, truncate it in the signal domain: $f_B = f\mathbf{1}_{[-B,B]}$. Then $\|f - f_B\|_{L^2} \to 0$ and

$$\|P_i f\|_{L^2} \le \|f - f_B\|_{L^2} + \|P_i f_B\|_{L^2}.$$

Again choose B first and then take $i \to -\infty$. This proves the coarse-resolution limit for every f. If $f \in \bigcap_i V_i$, then $P_i f = f$ at every resolution, so this limit forces $f = 0$.

What the proof has established

The normalized product defines a compactly supported, unit-energy scaling function. The box-seeded cascade converges to it in L^2, its integer translates are orthonormal, and the resulting nested spaces have both MRA limits. Together, these properties justify passing from repeated finite refinement steps to an orthonormal multiresolution representation.

A normalized correspondence. The product uniquely determines an integrable refinement solution with integral one. In the other direction, orthonormal coordinates recover the unique taps $g_0[n] = \langle \phi, \sqrt{2}\phi(2 \cdot -n)\rangle_{L^2}$. Multiplying a refinement solution by a nonzero constant leaves its filter unchanged; integral one removes this amplitude freedom.

11.8 Recovering the Haar scaling function from its filter

We now apply the theorem to Haar and recover the scaling function directly from the filter. This gives an explicit example of the construction just proved. Take

$$g_0[0] = g_0[1] = \frac{1}{\sqrt{2}}, \qquad G_0(z) = \frac{1 + z^{-1}}{\sqrt{2}}.$$

It satisfies the theorem. Its taps sum to $\sqrt{2}$, and

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})|^2 = 1 + \cos\omega,$$
$$|G_0(\mathrm{e}^{\mathrm{j}(\omega+\pi)})|^2 = 1 - \cos\omega.$$

Their sum is two. Also

$$|G_0(\mathrm{e}^{\mathrm{j}\omega})| = \sqrt{2}\cos(\omega/2) \ge 1 \qquad (|\omega| \le \pi/2).$$

The sufficient conditions are therefore satisfied. To identify the scaling function, we evaluate its infinite product. The normalized response is

$$m_0(\theta) = \frac{1}{2}(1 + \mathrm{e}^{-\mathrm{j}\theta}),$$

so the finite product is

$$\mathcal{P}_I(\omega) = \frac{1}{2^I} \prod_{i=1}^{I} (1 + \mathrm{e}^{-\mathrm{j}\omega/2^i}). \tag{11.15}$$

Why this product contains every dyadic phase once. From each factor choose either 1 or its exponential. A choice $(b_1, \ldots, b_I) \in \{0, 1\}^I$ contributes

$$\mathrm{e}^{-\mathrm{j}\omega \sum_{i=1}^{I} b_i 2^{-i}} = \mathrm{e}^{-\mathrm{j}\omega k/2^I}, \qquad k = \sum_{i=1}^{I} b_i 2^{I-i}.$$

Binary representation gives every integer $k = 0, \ldots, 2^I - 1$ exactly once. Hence

$$\boxed{\mathcal{P}_I(\omega) = \frac{1}{2^I} \sum_{k=0}^{2^I-1} \mathrm{e}^{-\mathrm{j}\omega k/2^I}.} \tag{11.16}$$

For example, with two factors, the four phases are

$$1, \quad \mathrm{e}^{-\mathrm{j}\omega/4}, \quad \mathrm{e}^{-\mathrm{j}\omega/2}, \quad \mathrm{e}^{-3\mathrm{j}\omega/4}.$$

This is the same finite-product identity that appeared in the tent-cascade calculation. There it described the heights of the dyadic cells; here it expresses the Fourier product as an average of complex exponentials. The identity also makes the limit accessible through a geometric sum.

Evaluate the geometric sum before passing to the limit. Set $M = 2^I$ and $r = \mathrm{e}^{-\mathrm{j}\omega/M}$. Since

$$(1 - r) \sum_{k=0}^{M-1} r^k = 1 - r^M,$$

we obtain, when $r \neq 1$,

$$\mathcal{P}_I(\omega) = \frac{1 - \mathrm{e}^{-\mathrm{j}\omega}}{M(1 - \mathrm{e}^{-\mathrm{j}\omega/M})}. \tag{11.17}$$

At exceptional frequencies, the finite sum (11.16) remains the definition. For each fixed $\omega \neq 0$, the quotient is valid for all sufficiently large I.

The first-order expansion $\mathrm{e}^{-\mathrm{j}\theta} = 1 - \mathrm{j}\theta + O(\theta^2)$ gives

$$M(1 - \mathrm{e}^{-\mathrm{j}\omega/M}) = \mathrm{j}\omega + O(\omega^2/M) \longrightarrow \mathrm{j}\omega.$$

Consequently,

$$\begin{aligned}\widehat{\phi}(\omega) &= \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega} = \mathrm{e}^{-\mathrm{j}\omega/2}\frac{\mathrm{e}^{\mathrm{j}\omega/2}-\mathrm{e}^{-\mathrm{j}\omega/2}}{\mathrm{j}\omega} \\ &= \mathrm{e}^{-\mathrm{j}\omega/2}\frac{\sin(\omega/2)}{\omega/2}, \qquad \omega \neq 0.\end{aligned} \tag{11.18}$$

At $\omega = 0$, every factor in (11.15) equals one. Thus $\widehat{\phi}(0) = 1$, the continuous extension of (11.18).

Identify the inverse transform. Direct integration gives

$$\int_0^1 \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t = \left[\frac{\mathrm{e}^{-\mathrm{j}\omega t}}{-\mathrm{j}\omega}\right]_{t=0}^{t=1} = \frac{1-\mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega}.$$

Therefore uniqueness of the Fourier transform yields

$$\boxed{\phi(t) = \mathbf{1}_{[0,1)}(t) \quad \text{as an } L^2 \text{ function.}} \tag{11.19}$$

Values at the two endpoints do not affect this statement. The phase factor $\mathrm{e}^{-\mathrm{j}\omega/2}$ locates the box at $[0,1]$, rather than centering it at zero.

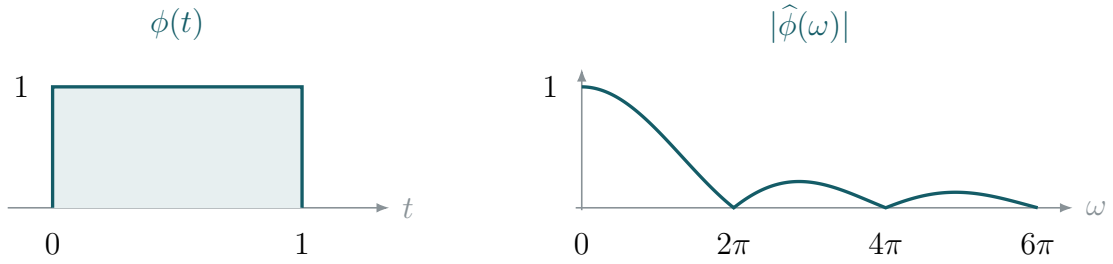


Figure 11.6: The Haar product recovers the unit box and its sinc-shaped Fourier magnitude. The right panel shows nonnegative frequencies; the magnitude is even.

11.9 Two interpretations of the Haar product

A Riemann sum. Equation (11.16) is the left-endpoint Riemann sum for

$$\int_0^1 \mathrm{e}^{-\mathrm{j}\omega t}\,\mathrm{d}t.$$

For fixed ω, the integrand is continuous. Thus the product-to-box calculation can also be understood as an increasingly fine average of complex exponentials on $[0,1]$.

Euler's cosine-product identity. Factoring each term as

$$\frac{1+\mathrm{e}^{-\mathrm{j}\theta}}{2} = \mathrm{e}^{-\mathrm{j}\theta/2}\cos(\theta/2)$$

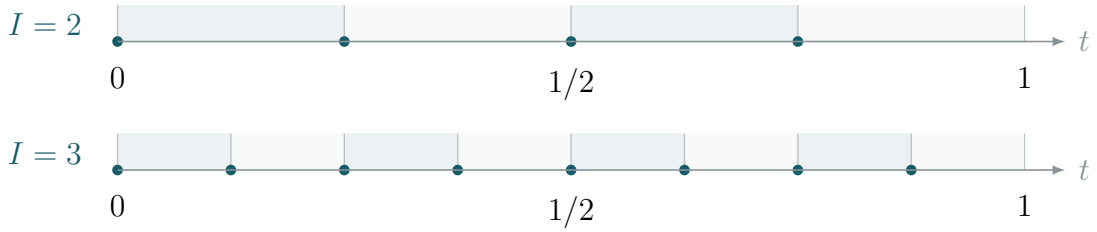


Figure 11.7: The 2^I choices in the Haar product select the left endpoints of the dyadic cells. Each term is weighted by the cell width 2^{-I}.

gives

$$\mathcal{P}_I(\omega) = \mathrm{e}^{-\mathrm{j}\omega \sum_{i=1}^{I} 2^{-(i+1)}} \prod_{i=1}^{I} \cos\left(\frac{\omega}{2^{i+1}}\right).$$

The geometric sum in the exponent is $(1 - 2^{-I})/2$. Comparing the limit with (11.18), and writing $x = \omega/2$, yields

$$\boxed{\frac{\sin x}{x} = \prod_{i=1}^{\infty} \cos\left(\frac{x}{2^i}\right),} \tag{11.20}$$

with value one at $x = 0$. This identity also follows by iterating the double-angle formula:

$$\sin x = 2^I \sin(x/2^I) \prod_{i=1}^{I} \cos(x/2^i), \qquad 2^I \sin(x/2^I) \longrightarrow x.$$

For fixed $x \neq 0$, division is valid for all sufficiently large I; then take the limit.

Both interpretations give the same result: successive dyadic refinements produce the Fourier transform of the unit box. The finite sum explains the sampling locations, while the cosine product separates the translation phase from the magnitude.

11.10 From the scaling function to an orthogonal wavelet

Suppose g_0 is supported in $\{0, \ldots, K\}$ with K odd. Form its conjugate-mirror partner and the associated function:

$$g_1[n] = (-1)^n g_0^*[K - n], \qquad \psi(t) = \sqrt{2} \sum_n g_1[n] \phi(2t - n).$$

Conjugation is needed for complex-valued taps. The same Fourier substitution used for ϕ gives

$$\boxed{\hat{\psi}(\omega) = \frac{G_1(\mathrm{e}^{\mathrm{j}\omega/2})}{\sqrt{2}} \hat{\phi}(\omega/2).} \tag{11.21}$$

The discrete orthonormal basis. We first recall the sequence-space argument from Lecture 10 so that we can transfer each of its conclusions to functions. Define the vectors and correlations

$$v_{\ell,k}[n] = g_\ell[n-2k], \qquad r_{\ell s}[q] = \sum_n g_\ell[n] g_s^*[n-2q].$$

The low-pass condition is $r_{00}[q] = \delta[q]$. Substitution in the mirror rule gives

$$r_{11}[q] = \sum_n g_0^*[K-n] g_0[K-n+2q] = r_{00}[q] = \delta[q],$$
$$r_{01}[q] = \sum_n (-1)^n g_0[n] g_0[K+2q-n] = 0.$$

For the last equality, pair n with $K+2q-n$: the product is unchanged, but its sign reverses because $K+2q$ is odd. Also $r_{10}[q] = r_{01}^*[-q] = 0$. Since $\langle v_{\ell,k}, v_{s,m}\rangle_{\ell^2} = r_{\ell s}[m-k]$, the family is orthonormal.

For completeness, let $e_r[n] = \delta[n-r]$ and form its finite projection $p_{r,N} = \sum_{\ell=0}^{1} \sum_{k=-N}^{N} \langle e_r, v_{\ell,k}\rangle_{\ell^2} v_{\ell,k}$. Expanding its squared error using orthonormality gives

$$0 \le \|e_r - p_{r,N}\|_{\ell^2}^2 = 1 - \sum_{\ell=0}^{1} \sum_{k=-N}^{N} |g_\ell[r-2k]|^2.$$

Thus $S_r = \sum_{\ell,k} |g_\ell[r-2k]|^2 \le 1$. The even and odd indices partition both filters, so $S_0 + S_1 = \|g_0\|_{\ell^2}^2 + \|g_1\|_{\ell^2}^2 = 2$. It follows that $S_0 = S_1 = 1$. Reindexing gives $S_{r+2} = S_r$, hence every $S_r = 1$ and $p_{r,N} \to e_r$. Every unit impulse is therefore in the closed span. Since finite truncations of any ℓ^2 sequence converge in norm, that closed span is all of $\ell^2(\mathbb{Z})$.

Transfer from sequences to functions. The discrete basis result becomes a statement about functions because the translates of the scaling function already form an orthonormal basis of V_1. In particular, using these functions as synthesis atoms identifies a coefficient sequence with a function without changing its norm. The map

$$c \longmapsto \sum_n c[n] \phi_{1,n}, \qquad \phi_{1,n}(t) = \sqrt{2}\phi(2t-n),$$

preserves the norm: for finite sums, orthonormality gives squared norm $\sum_n |c[n]|^2$, and passage to the limit gives the same identity for $c \in \ell^2$. Its range is V_1. Under this map, the two discrete families become

$$\phi(t-k) = \sum_n g_0[n-2k] \phi_{1,n}(t), \qquad \psi(t-k) = \sum_n g_1[n-2k] \phi_{1,n}(t).$$

For example,

$$\begin{aligned}\langle \psi(\cdot - k), \psi(\cdot - m)\rangle_{L^2} &= \sum_{n,r} g_1[n-2k] g_1^*[r-2m] \langle \phi_{1,n}, \phi_{1,r}\rangle_{L^2} \\ &= \sum_{n} g_1[n-2k] g_1^*[n-2m] = \delta[k-m].\end{aligned}$$

Replacing the first g_1 by g_0 gives the cross inner product zero. The discrete completeness just proved transfers through the norm-preserving map. Hence

$$V_1 = V_0 \oplus W_0, \qquad W_0 = \overline{\operatorname{span}\{\psi(\cdot - n) : n \in \mathbb{Z}\}}^{L^2}.$$

Dilation gives $V_{i+1} = V_i \oplus W_i$, where $W_i = \overline{\operatorname{span}\{\psi_{i,n} : n \in \mathbb{Z}\}}^{L^2}$ and $\psi_{i,n}(t) = 2^{i/2}\psi(2^i t - n)$. If $r < i$, then $W_r \subset V_{r+1} \subset V_i$ while $W_i \perp V_i$; thus different detail spaces are orthogonal.

To establish completeness, it is the span of the detail spaces that must be considered, not their set-theoretic union. The orthogonal projection onto W_i is $Q_i = P_{i+1} - P_i$. Thus, for integers $A < B$, telescoping gives

$$\sum_{i=A}^{B-1} Q_i f = P_B f - P_A f.$$

The two MRA limits proved above give $P_B f \to f$ as $B \to \infty$ and $P_A f \to 0$ as $A \to -\infty$. Thus the sum of the detail projections converges to f in L^2. Consequently,

$$\boxed{L^2(\mathbb{R}) = \overline{\bigoplus_{i\in\mathbb{Z}} W_i}^{L^2}, \qquad \{\psi_{i,n} : i, n \in \mathbb{Z}\} \text{ is an ONB.}} \tag{11.22}$$

The direct sum means finite sums of elements of mutually orthogonal detail spaces; its closure includes their L^2 limits.

11.11 Completing the Haar example

For Haar, $K = 1$ and the conjugate-mirror convention gives

$$g_1[0] = \frac{1}{\sqrt{2}}, \qquad g_1[1] = -\frac{1}{\sqrt{2}}.$$

Substituting the recovered scaling function in its two-scale expansion,

$$\begin{aligned}\psi(t) &= \sqrt{2}\left(\frac{1}{\sqrt{2}}\phi(2t) - \frac{1}{\sqrt{2}}\phi(2t-1)\right) \\ &= \mathbf{1}_{[0,1/2)}(t) - \mathbf{1}_{[1/2,1)}(t).\end{aligned}$$

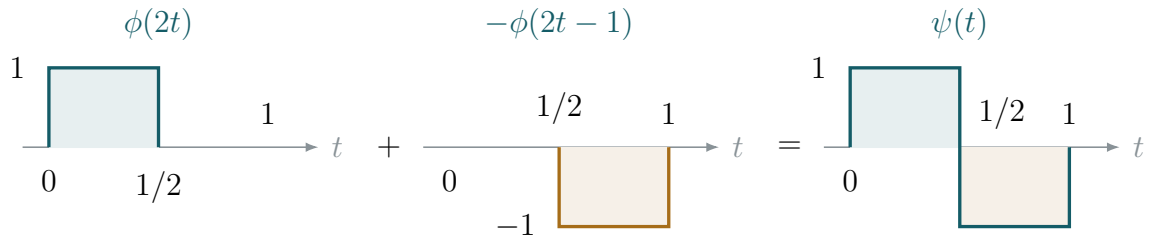


Figure 11.8: The high-pass taps convert the recovered scaling function into the Haar wavelet. The positive and negative halves have equal area, so $\int_{\mathbb{R}} \psi(t)\,\mathrm{d}t = 0$.

The low-pass taps add two half-width boxes; the high-pass taps subtract them.

The sign of the partner is a convention. Replacing g_1 by $-g_1$ replaces ψ by $-\psi$. Its detail coefficients also change sign, so each synthesis term $\langle f, \psi_{i,n}\rangle_{L^2}\psi_{i,n}$ is unchanged. We use the same sign as Lecture 10.

For Haar, the opposite-sign convention can be written

$$\tilde{g}_1[n] = (-1)^{1-n} g_0^*[1-n], \qquad \big(\tilde{g}_1[0], \tilde{g}_1[1]\big) = \frac{1}{\sqrt{2}}(-1, 1).$$

It gives the negative of the displayed Haar wavelet. Thus a negative first half and a positive second half describe the same orthonormal construction, with the signs of the detail coefficients changed accordingly.

A final check on the tent. For the three-tap filter,

$$m_0(\theta) = \frac{1}{4}(1 + \mathrm{e}^{-\mathrm{j}\theta})^2 = m_{\mathrm{Haar}}(\theta)^2.$$

Its limiting product is therefore

$$\hat{\tau}(\omega) = \left(\frac{1 - \mathrm{e}^{-\mathrm{j}\omega}}{\mathrm{j}\omega}\right)^2 = \mathrm{e}^{-\mathrm{j}\omega}\left(\frac{\sin(\omega/2)}{\omega/2}\right)^2.$$

The value at $\omega = 0$ is one. The convolution of two unit boxes at t is the length of $[0,1] \cap [t-1, t]$: this is t for $0 \le t \le 1$, $2 - t$ for $1 \le t \le 2$, and zero outside. Thus $\tau = \mathbf{1}_{[0,1)} * \mathbf{1}_{[0,1)}$, agreeing with the cascade calculation.

The construction has now connected the three descriptions of a wavelet system. The filter specifies a refinement rule, the infinite product identifies its normalized limit, and the MRA properties ensure that the resulting wavelets give a complete orthonormal representation. The hypotheses on the filter are what make these descriptions consistent.